

Note on Lagrangian Method in Macroeconomics

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1 Continuous Growth Model

Household's Problem:

$$U = \int_0^{\infty} e^{-\rho t} \log C(t) dt \quad (1)$$

subject to

$$\begin{aligned} \dot{k}(t) &= f(k) - c(t) - \delta k(t) \\ f(k) &= k(t)^\alpha \\ k(t) &> 0 \end{aligned}$$

The Lagrangian function is:

$$L = \int_0^{\infty} e^{-\rho t} [\log c(t) + \lambda(t)(-\dot{k}(t) + k(t)^\alpha - c(t) - \delta k(t))] dt \quad (2)$$

$\dot{k}(t)$ is annoying here, so we need to find some way to dissolve it. One possible solution is chain-rule.

$$\int f'(x)g(x)dx = f(x)g(x) - \int f(x)g'(x)dx \quad (3)$$

So we have:

$$L = \int_0^{\infty} e^{-\rho t} [\log c(t) + -\lambda(t)\dot{k}(t) + \lambda(t)(k(t)^\alpha - c(t) - \delta k(t))] dt - e^{-\rho t} \lambda(t)k(t)|_0^{\infty} \quad (4)$$

Transversality Condition:

$$\lim_{t \rightarrow \infty} e^{-\rho t} \lambda(t)k(t) = 0 \quad (5)$$

The first order condition:

$$\begin{cases} \frac{\partial L}{\partial c} = e^{-\rho t} \left(\frac{1}{c(t)} - \lambda(t) \right) = 0 \\ \frac{\partial L}{\partial k} = e^{-\rho t} (\lambda(t) - \rho \lambda(t) + \lambda(t) \alpha k(t)^{\alpha-1} - \lambda(t) \delta) = 0 \end{cases} \quad (6)$$

So the economic system can be described by:

$$\begin{cases} \frac{\dot{c}(t)}{c(t)} = \alpha k(t)^{\alpha-1} - \rho - \delta \\ k(t) = f(k) - c(t) - \delta k(t) \\ \lim_{t \rightarrow \infty} e^{-\rho t} \lambda(t) k(t) = 0 \end{cases} \quad (7)$$

2 Discrete RBC Model

Brock and Mirman (1972) make the growth model stochastic, by positing that the production technology is buffeted by random aggregate shocks to productivity.

2.1 The structure

- Preferences

$$E_0 \sum_{t=0}^{\infty} b^t u(C_t, L_t), \quad b > 0, \quad (8)$$

where b denotes the discount factor, C_t represent consumption and L_t leisure. The momentary utility function $u(C_t, L_t)$ is assumed concave and obeys regularity conditions.

- Endowments

$$N_t + L_t = 1 \quad (9)$$

- Technology The production function

$$Y_t = A_t F(K_t, N_t X_t) \quad (10)$$

is constant returns to scale, where A_t is a random “productivity shock” variable, and X_t represents the deterministic component of productivity. Furtherly, it is assumed that:

$$X_{t+1} = \gamma X_t, \quad \gamma > 1. \quad (11)$$

- Constraints

$$Y_t = C_t + I_t \quad (12)$$

$$K_{t+1} = I_t + (1 - \delta)K_t \quad (13)$$

- Initial Conditions

$$K_0 > 0, \quad X_0 > 0, \quad A_0 > 0. \quad (14)$$

2.2 Steady-state growth and transforming the economy

In order to make a steady-state path desirable, in the standard fixed labor¹ version of the basic neoclassical model momentary utility has to take the form

$$u(C) = \frac{1}{1-\sigma} [C^{1-\sigma} - 1] \quad (15)$$

where $\sigma > 0$. This utility function insures that the marginal rate of substitution between consumption at dates t and $t+1$ depends only on the growth rate of consumption.

When these conditions are imposed, we can transform the economy by scaling all the trending variables by the initial level of X , to eliminate the steady-state growth.

So the optimal growth problem can be written as:

$$\sum_{t=0}^{\infty} \beta^t u(c_t, L_t) \quad (16)$$

where $\beta = b\gamma^{1-\sigma}$, satisfying $0 < \beta < 1$. Utility is maximized subject to the transformed constraints:

$$\begin{cases} N_t = 1 - L_t \\ y_t = A_t F(k_t, N_t) \\ y_t = c_t + i_t \\ \gamma k_{t+1} = i_t + (1 - \delta)k_t \end{cases} \quad (17)$$

2.3 Optimal capital accumulation

The Lagrangian function is:

$$L = \sum_{t=0}^{\infty} \beta^t \{u(c_t, L_t) + \lambda_t [A_t F(k_t, N_t) + (1 - \delta)k_t - c_t - \gamma k_{t+1}] + \omega_t [1 - L_t - N_t]\} \quad (18)$$

The F.O.C are:

$$\begin{cases} c_t & : D_1 u(c_t, L_t) = \lambda_t, \\ L_t & : D_2 u(c_t, L_t) = \omega_t, \\ N_t & : \lambda_t A_t D_2 F(k_t, N_t) = \omega_t, \\ k_{t+1} & : \beta \lambda_{t+1} [A_{t+1} D_1 F(k_{t+1}, N_{t+1}) + 1 - \delta] = \gamma \lambda_t. \end{cases} \quad (19)$$

An optimal consumption, leisure, work and capital plan- sequences $\{c_t\}_{t=0}^{\infty}$, $\{L_t\}_{t=0}^{\infty}$, $\{N_t\}_{t=0}^{\infty}$, and $\{k_t\}_{t=0}^{\infty}$ - satisfies these first order conditions, the original constraints, the initial condition

¹For endogenous labor supply situation,

$$u(c, L) = \frac{1}{1-\sigma} ([C\mu(L)]^{1-\sigma} - 1)$$

is required.

requirement on k_t and the transversality condition, $\lim_{t \rightarrow \infty} \beta^t \lambda_t k_{t+1} = 0$.

Optimal capital accumulation in the basic neoclassical model is a “general equilibrium” phenomenon in three ways:

- The choices of consumption, labor and capital accumulation are interdependent at each point in time and across time;
- The requirement that the optimal decisions must respect the resource constraints of the economy is signalled by shadow prices;
- If these shadow prices were market prices for individual households, then they would similarly signal these agents to supply and demand the optimal quantities.

2.4 The nature of the steady state

When $A_t = A$ for all t , we can get a unique stationary state:

- From the capital accumulation efficiency condition,

$$AD_1 F(k, N) = (r + \delta) \quad (20)$$

where, $r = \frac{\gamma}{\beta} - 1$ is the stationary state real interest rate. As $F(k, N)$ is constant returns to scale, equation (20) determines the capital-labor ratio as a function of productivity and the rental rate.

- The real wage $w = \frac{\omega}{\lambda}$ is determined by:

$$w = \frac{\omega}{\lambda} = AD_2 F\left(\frac{k}{N}, 1\right) \quad (21)$$

- There are unique levels of work, consumption and the shadow price of consumption that satisfy the remaining equations.

Hence, the original economy will be in the steady state with many variables-including consumption, capital output and real wages- growing at the same rate.

References

- King, Robert, and Sergio, Rebelo (1999), Resuscitating Real Business Cycles, Chapter 14 of *Handbook of Macroeconomics*, Elsevier, Volume 1, Part B, Pages 927-1007.
- King, Robert, Charles, Plosser, and Sergio, Rebelo (2002) Production, Growth and Business Cycles: Technical Appendix, *Computational Economics* 20: 87-116.